

AISHIK KUMAR SARKAR

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PROFESSIONAL EXPERIENCE

ST ENGINEERING | *AI Engineer*

MAY 2026 – PRESENT

- Cut manual documentation lookup time by 70% by architecting and deploying a self-hosted RAG platform, enabling maintenance engineers to semantically search 500+ page manuals and design docs in natural language.
- Enabled prompt orchestration across 5+ AI models by re-engineering IBM OpenRAG's authentication, request-routing, and LangFlow pipelines, integrating OpenRouter and OpenSearch to support flexible, multi-model AI workflows across the platform.
- Cut inference costs by 60% by migrating the platform from cloud-based APIs to 100% local GPU inference using vLLM, eliminating external API dependency while improving response latency and strengthening data privacy.
- Unlocked 3+ new automated workflows by integrating Model Context Protocol (MCP) servers for Excel, Word, and MongoDB into the platform, extending its capabilities beyond document search into structured data automation.

KAVEMAN PRODUCTIONS | *AI Software Engineer*

MAR 2026 – MAY 2026

- Unified fragmented project tracking, admin, and client coordination into a single Project & Workflow Management SuperApp with real-time sync, built with React/Next.js, Supabase, and Tailwind CSS.
- Took core modules – task/project tracking, admin pipeline, client/partner portal, role-based permissions – from prototype to deployed build within an 8-week Agile sprint cycle.
- Reduced rework risk by validating requirements through click-through Figma prototypes and stakeholder workshops before development began, translating creative-production workflows into technical specs.
- Improved cross-functional visibility into resource allocation and financial tracking by auditing system architecture and workflow gaps, then redesigning end-to-end processes and delivering full technical documentation plus a stakeholder presentation.

MECATRON ROBOTICS | *Full Stack Web Developer & Software Engineer*

JUL 2025 – MAY 2026

- Increased visibility for projects, events, and outreach by building and maintaining the official team website and internal tools for a 30+ member interdisciplinary engineering team.
- Increased user engagement and navigation efficiency by 40% by shipping dynamic front-end features using ReactJS, HTML, CSS, and JavaScript.
- Established responsive design and cross-browser testing practices, ensuring consistent site performance across desktop and mobile for prospective members and event attendees.
- Improved data retrieval performance by 35% by engineering back-end services, APIs, and database integrations managing 100+ content entries (projects, events, workshops).

SINGTEL – NCS GROUP | *Business & Data Analyst*

OCT 2021 – APR 2022

- Informed social programs benefiting 200+ families by analyzing and visualizing data using SQL, Excel, and SAS, then presenting insights to management.
- Improved query efficiency and reporting accuracy by maintaining and optimizing the team's data warehouse.
- Secured sensitive datasets by developing encryption and decryption workflows for confidential data handling.

SINGAPORE ARMED FORCES | *Cyber Security Operator (National Service)*

OCT 2022 – AUG 2024

- Ensured early detection of cyber-attacks by monitoring and analyzing network security threats on a shift-work basis.
- Strengthened incident-response readiness by mentoring 15+ personnel in response protocols and investigating security breaches.

KEY SKILLS

AI / ML / LLM Tooling: Retrieval-Augmented Generation (RAG) | LangFlow | OpenRAG | OpenSearch | vLLM | Model Context Protocol (MCP) | Gemini | Claude | Prompt Engineering

Programming Languages: Python | Java | C++ | C | JavaScript | SQL (MySQL) | HTML5 | CSS3 | Git

Frameworks & Platforms: ReactJS | Next.js | NodeJS | Supabase | Firebase

Familiar Software: Android Studio | Figma | IntelliJ IDEA | KNIME | PyCharm | Visual Studio Code

Spoken Languages: English | Bengali

HACKATHON PROJECTS

ReUnite – AI Lost & Found Platform | [\[NTU SummerBuild 2025 Winner – Most Useful\]](#)

MAY 2025 – JUL 2025

- Won the "Most Useful Award" out of 50+ teams, with potential adoption at NTU, by leading a 5-member team to modernize NTU's outdated lost-and-found system.
- Shipped a full-stack platform using ReactJS, NodeJS, and Firebase, validated with 40+ item listings.
- Boosted match accuracy by 60% by integrating Gemini AI for image-to-text description and visual similarity search.
- Cut notification latency to under 5 seconds by automating real-time alerts via EmailJS & Nodemailer.

Fact Checker – AI Fake News Detector | [\[NTU TechFest 2025\]](#)

MAR 2025 – MAR 2025

- Directed a 5-member team to build a Python-based Telegram bot that identified fake news articles.
- Improved news-credibility detection accuracy by 20% by applying prompt engineering with the Gemini API.
- Cut API response time by 30% by tuning requests, enabling smooth real-time interaction.

EDUCATION

NANYANG TECHNOLOGICAL UNIVERSITY | *B.Comp. (Hons.), Computer Science*

AUG 2024 – EXPECTED JUN 2028

- Double Specialisation in Artificial Intelligence & Data Science
- Upgraded to the Work-Study Degree Programme with ST Engineering (Industry Partner)
- Relevant Coursework: Data Structures & Algorithms | Artificial Intelligence | Software Engineering | Object Oriented Programming & Design | Operating Systems

SINGAPORE POLYTECHNIC | *Diploma in Computer Engineering*

AUG 2019 – APR 2022

- Singtel Engineering Scholar | CCA Gold with Honours Award | Certificate of Achievement 2020 | Director's Honour Roll 2019
- Relevant Coursework: Machine Learning & Artificial Intelligence | Client-Server Application Development | Fundamentals of Internet of Things | Embedded Computer Systems

LEADERSHIP & CO-CURRICULAR ACTIVITIES

NTU DEEP LEARNING WEEK (DLW) 2027 | *Deputy President*

MAY 2026 – PRESENT

- Lead strategic planning and execution for one of ASEAN's largest student-run AI & ML hackathons, engaging 1000+ participants through hackathons, workshops, and industry talks.
- Oversee 5 executive divisions (Academics, Digital Operations, Event Operations, Marketing, Corporate Relations), coordinating 50+ student organizers and cross-functional committees to deliver a week-long event.
- Drive partnerships with industry sponsors, universities, and AI professionals while overseeing hackathon strategy, judging workflows, and participant experience.

NTU MACHINE LEARNING & DATA ANALYTICS (MLDA) | *Community Projects Division*

AUG 2025 – MAY 2026

- Organized 3 technical events and 3+ workshops engaging 80+ attendees across ML and data analytics topics, coordinating logistics and external collaborations.
- Designed bonding activities and mini hackathons for 50+ active members, strengthening community engagement.

NTU DEEP LEARNING WEEK (DLW) 2026 | *Academics Division*

AUG 2025 – MAY 2026

- Developed the academic framework for a 7-day AI hackathon with 1000+ participants.
- Improved clarity and submission quality by defining technical scope, judging criteria, and evaluation workflow per track.
- Cut manual evaluation time and increased scoring accuracy by building an automated scoring system.